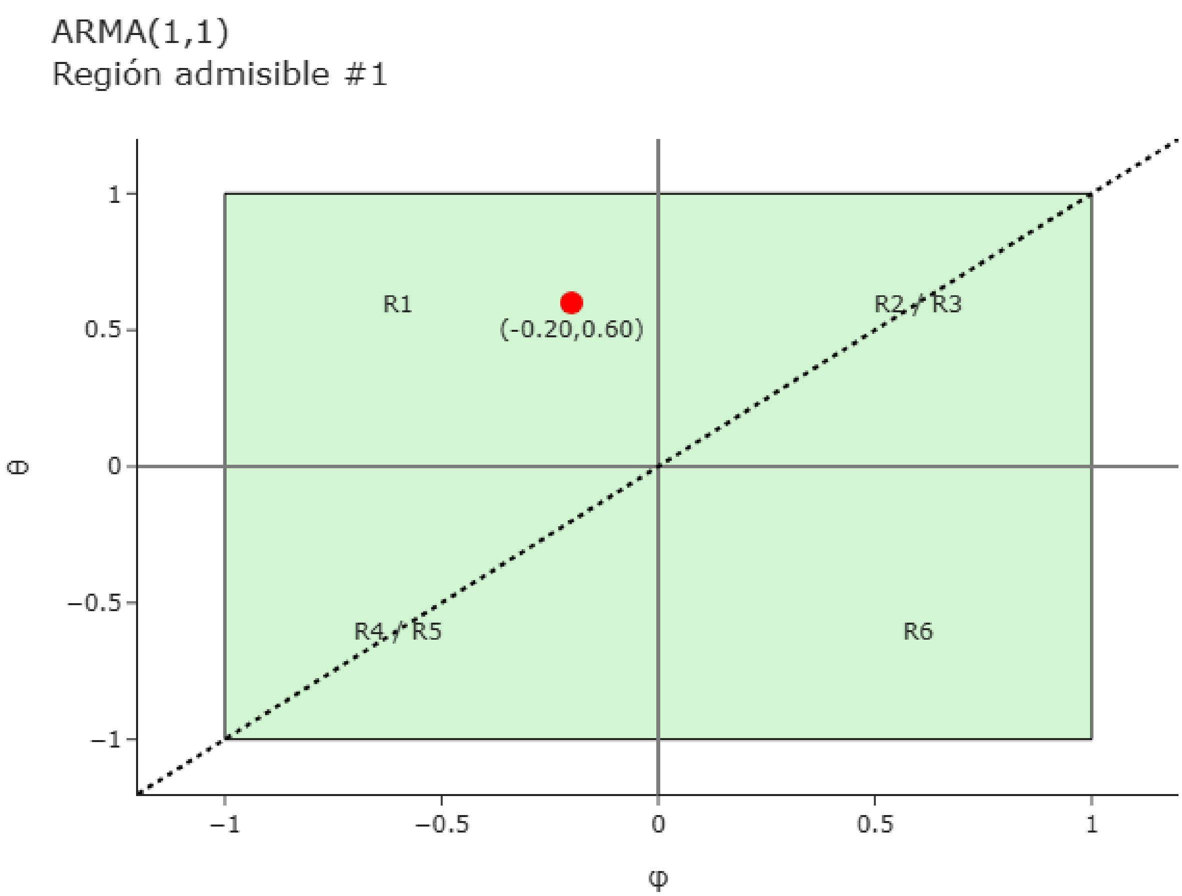
Regiones admisibles del proceso ARMA(1,1)

Considere un proceso ARMA(1,1):

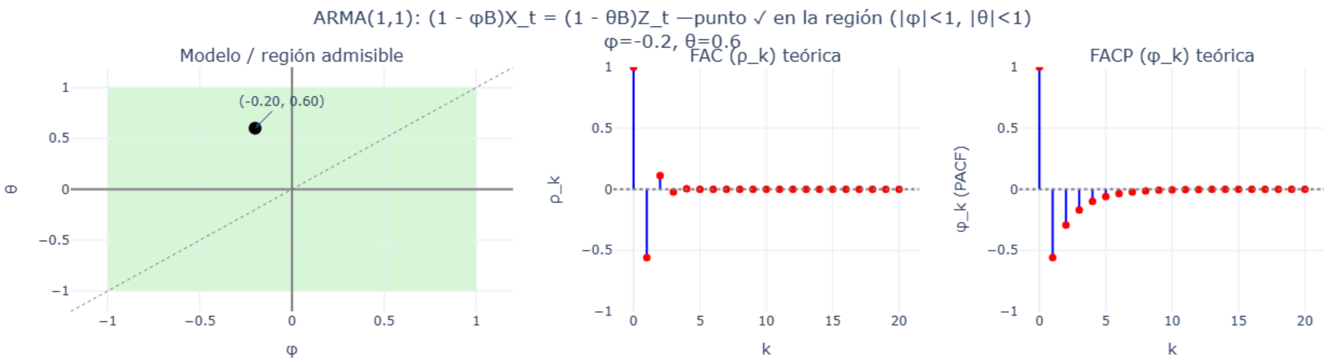
$$(1-\phi B)W_t = (1-\theta B)\alpha_t \quad ; \quad \alpha_t \sim \text{wn}(0, \sigma_\alpha^2)$$

A partir de las condiciones para estacionariedad e invertibilidad:
$|\phi| < 1$ y $|\theta| < 1$, se tienen las siguientes **regiones admisibles**:

Región admisible 1:
$-1 < \phi < 0$ y $0 < \theta < 1$



https://nuriaarroyo.github.io/ecoII/interactive/ARMA_1_1_br_Regi%C3%B3n_admisible_1.html



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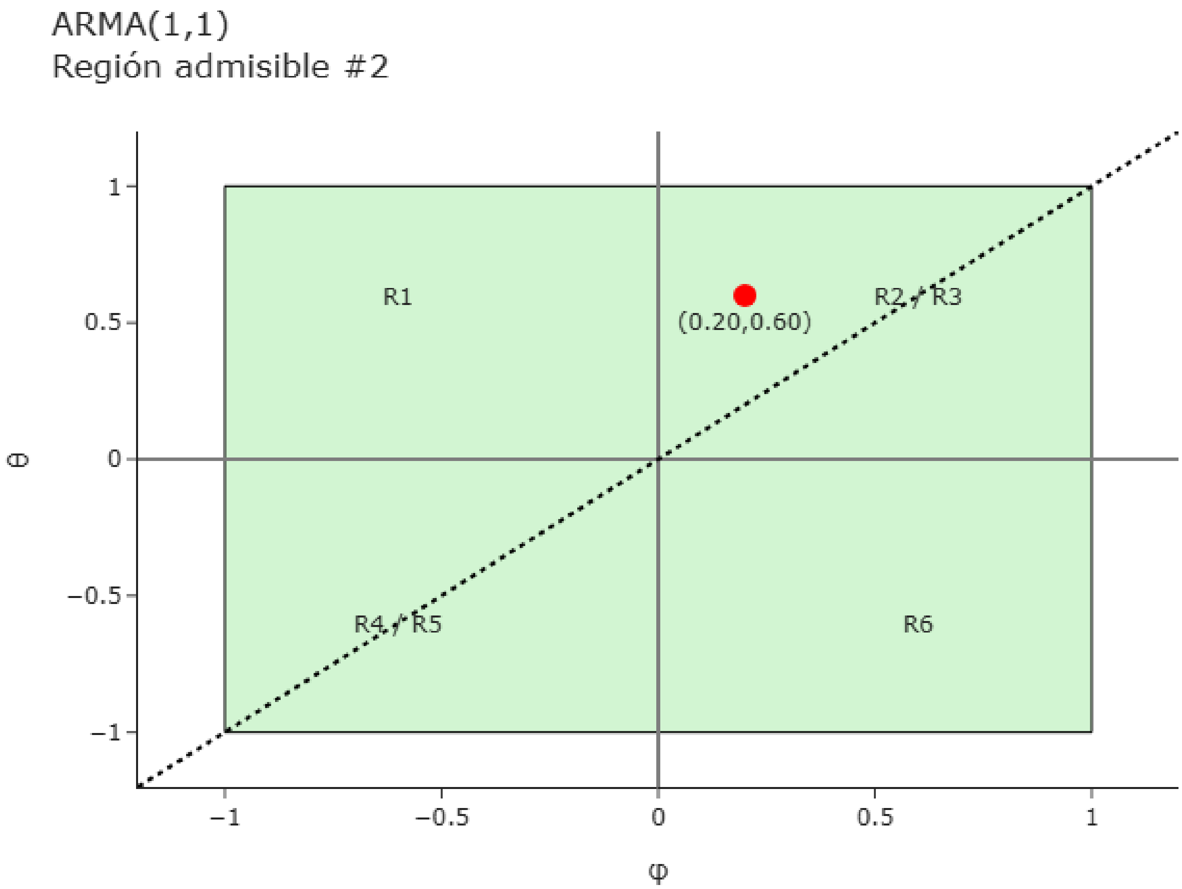
FAC teórica:

$$\rho_k = \frac{\phi^{k-1}(1-\theta)(\phi-\theta)}{1-2\phi\theta+\theta^2}, \quad k = 1, 2, \ldots$$

Región admisible 2:

$$0 < \phi < 1 \quad ; \quad 0 < \theta < 1 \quad ; \quad \theta > \phi$$



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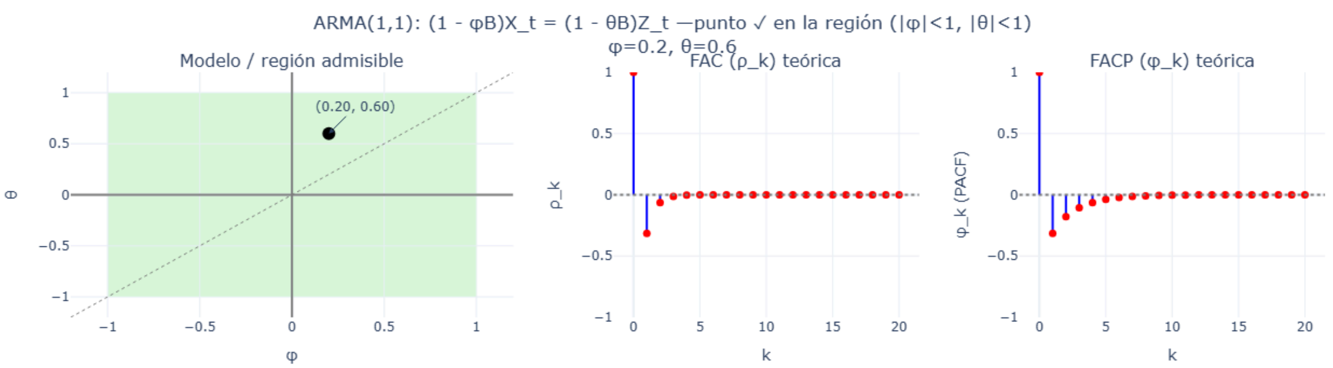

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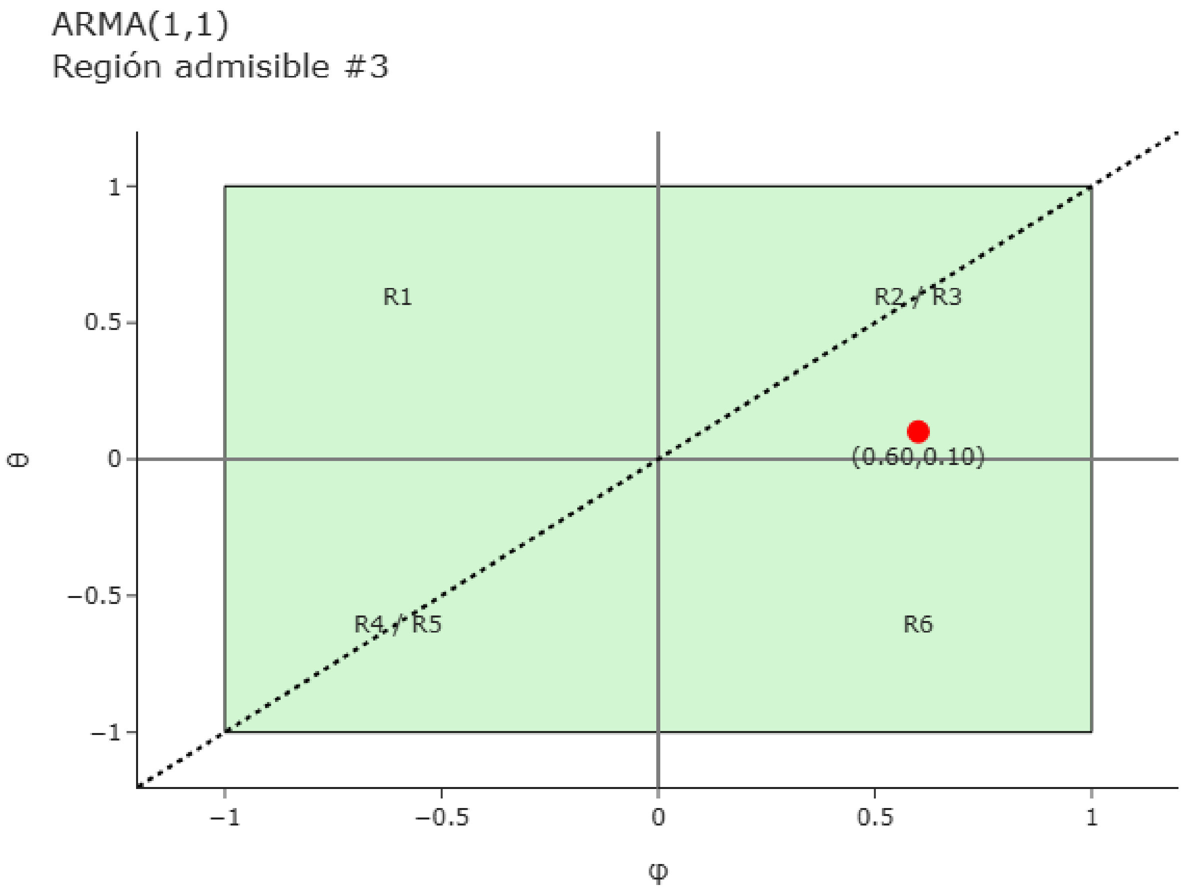
Región admisible 3:

$$0 < \phi < 1 \quad ; \quad 0 < \theta < 1 \quad ; \quad \theta < \phi$$



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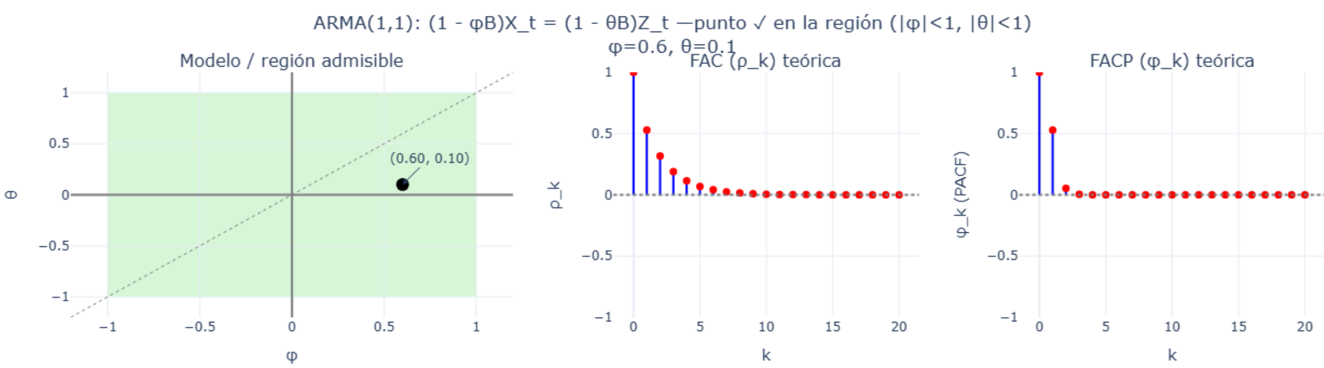

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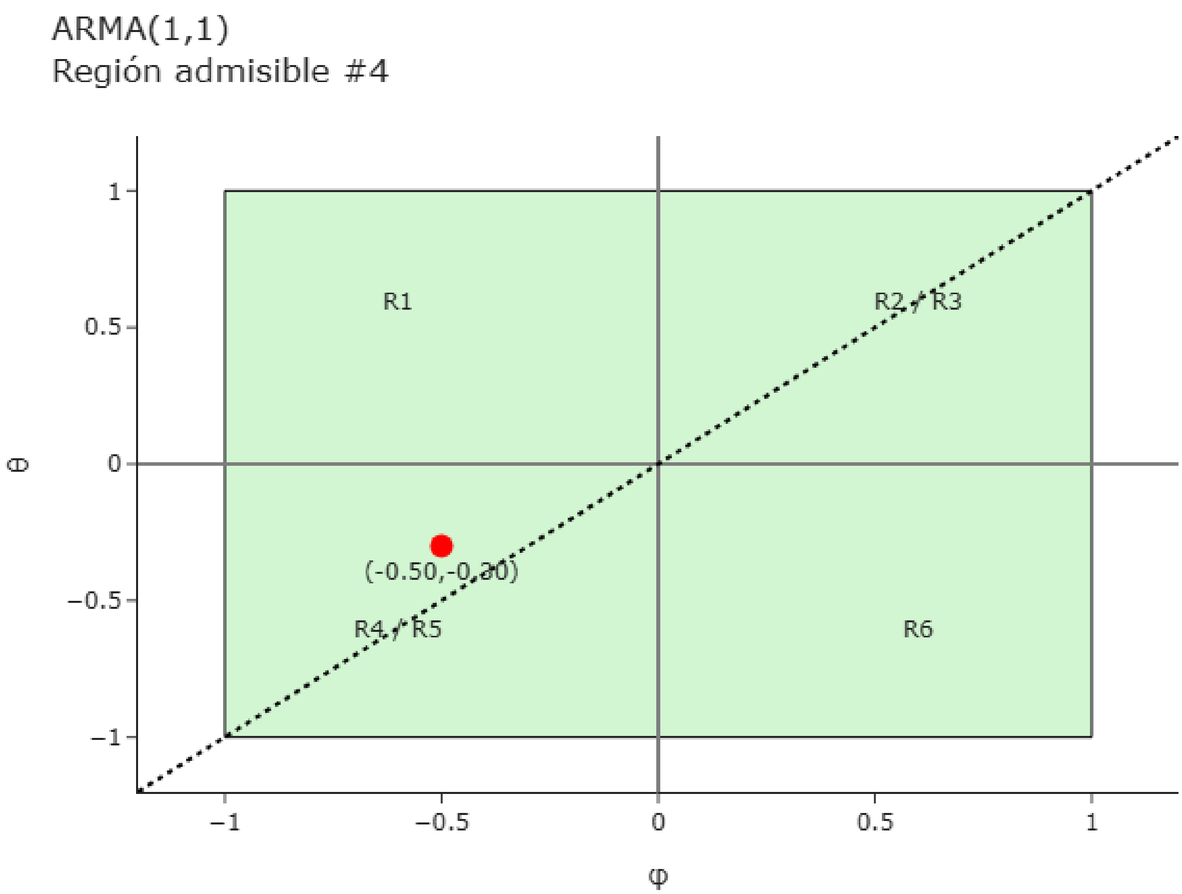
Región admisible 4:

$$-1 < \phi < 0 \quad ; \quad -1 < \theta < 0 \quad ; \quad \theta > \phi$$



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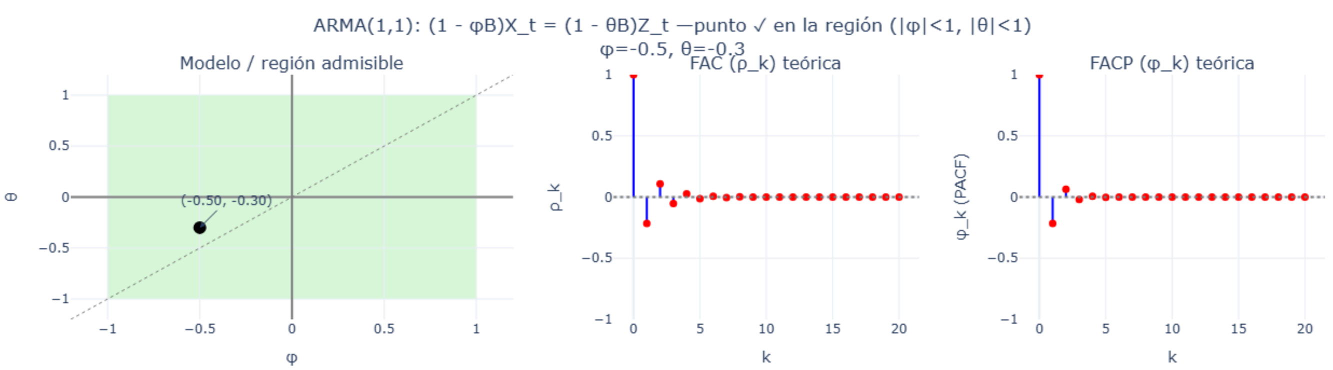

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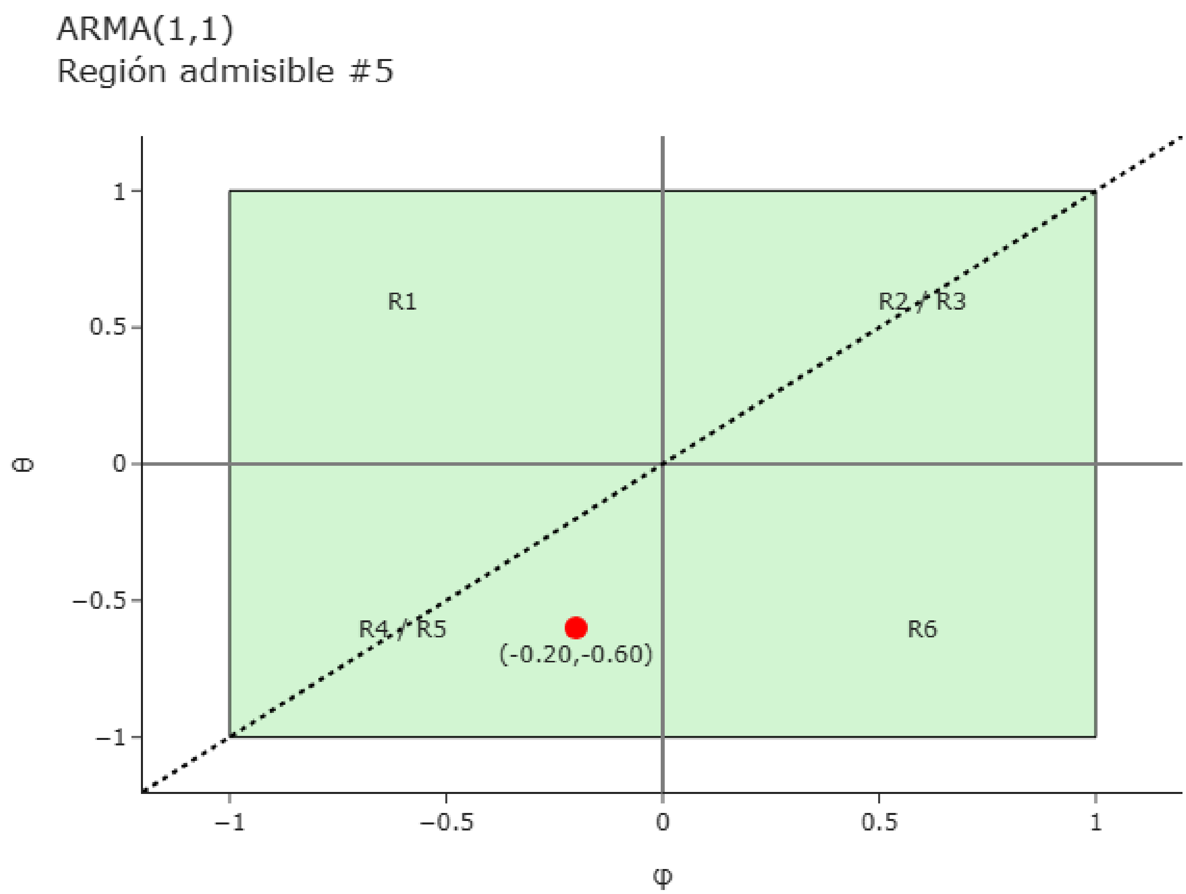
Región admisible 5:

$$-1 < \phi < 0 \quad ; \quad -1 < \theta < 0 \quad ; \quad \theta < \phi$$



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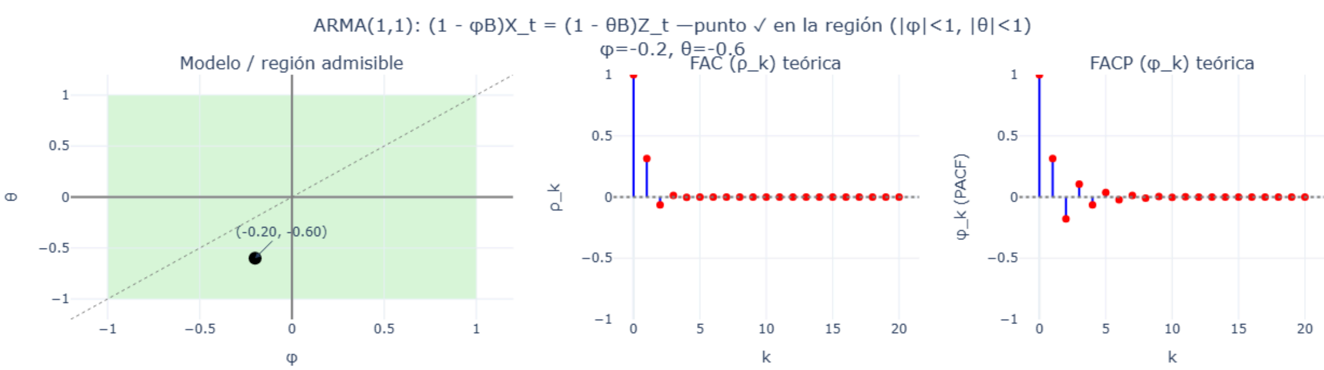

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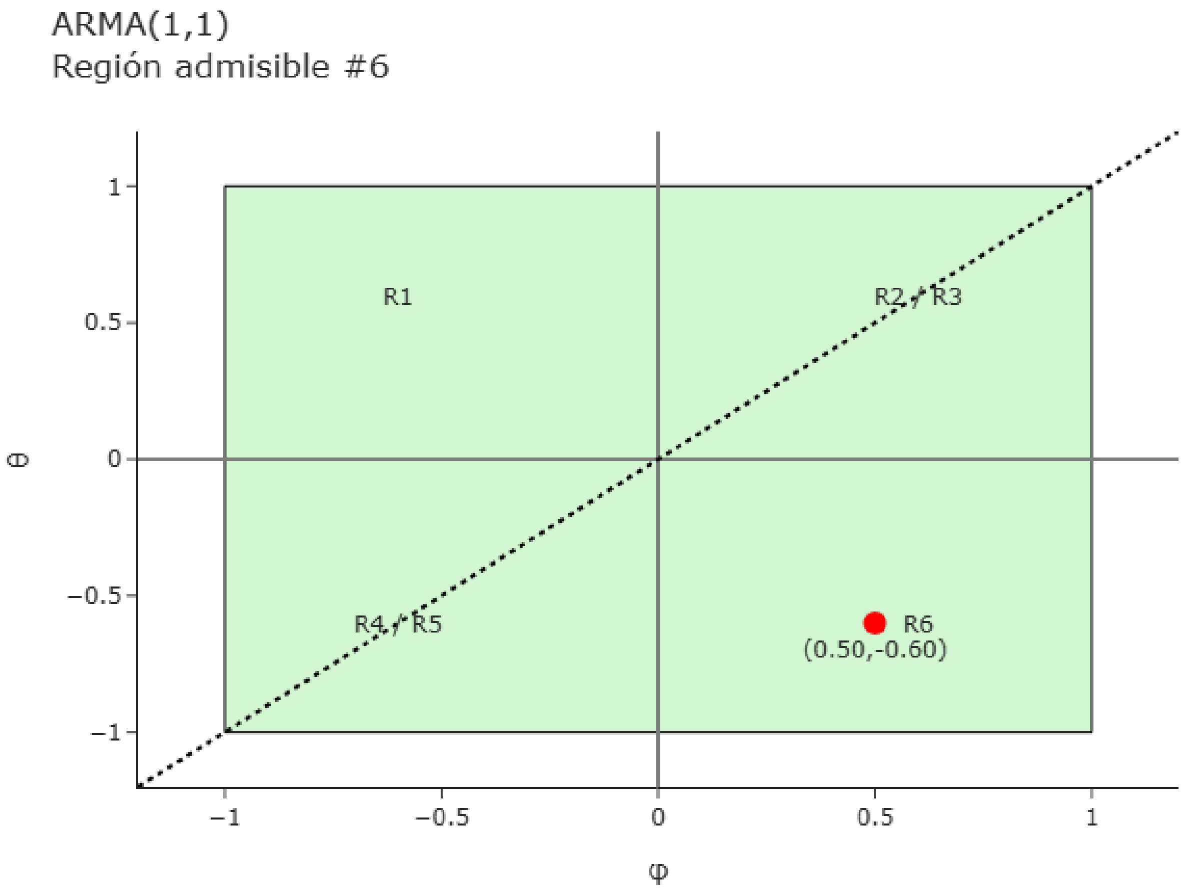
Región admisible 6:

$$0 < \phi < 1 \quad ; \quad -1 < \theta < 0$$



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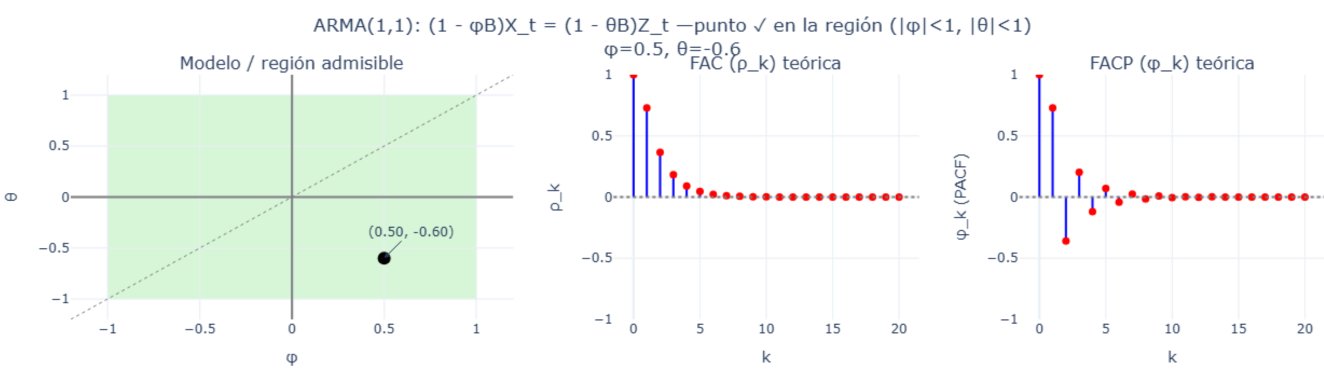

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